Object Localization in 3D Semantic Maps via Cluster Bearing and Monocular Distance

Author Name
Affiliation
Organization
City, Country
email address

***Abstract*—Localizing objects in a 3D semantic map for indoor autonomous delivery robots is commonly solved using stereo cameras, RGB-D or LiDAR sensors, or camera-based triangulation methods that depend on the pose odometry of each frame. This paper presents an alternative approach that uses only a low-cost monocular camera (ESP32-CAM), where accuracy comes from algorithm design rather than hardware upgrades. The bearing of the target object is computed from a semantic point cluster — MapAnything reconstructs the point cloud, YOLO11n attaches the class label, and DBSCAN performs clustering — with the result transformed into real-world coordinates using the Umeyama algorithm, fitted once over the entire camera trajectory. Distance is estimated from a single frame using the YOLO bounding-box height together with basic geometric principles. Experiments were conducted over sixteen independent capture sessions at three ground-truth distances. At $3\,\mathrm{m}$ and $5\,\mathrm{m}$ the mean error stays under 2%, with standard deviations of $0.19\,\mathrm{m}$ and $0.22\,\mathrm{m}$; at $7\,\mathrm{m}$ a systematic bias of about -11% appears, but the standard deviation is only $0.07\,\mathrm{m}$, indicating that the error is dominated by a calibratable bias rather than random noise.**

***Index Terms*—3D semantic map, object localization, MapAnything, YOLO, autonomous delivery robot, global fit Umeyama, monocular distance estimation**

I. Introduction

The indoor autonomous delivery robot needs to construct and maintain a semantic 3D map to understand object localization and reference points— the object to be delivered, as well as fixed objects such as plant pots, tables, and chairs that can serve as reference points for navigation. A typical pipeline that combines three components : (i) a 2D detection model (YOLO) that runs on each frame, (ii) a monocular 3D reconstruction model, and (iii) a mechanism for determining the absolute position of objects in the real world, often depends on information from localizing odometry of the robot.

The real-world deployment with low-cost hardware and odometry that accumulates errors over time raises a fundamental question: what really determines the accuracy, and what can we do to improve confidence under limited resources? On the robot Hawkbot (ROS2 Jazzy) equipped with the camera ESP32-CAM, we realize that methodologies that depend on pose odometry of every single frame are sensitive to local noise. The paper presents another methodology: combining orientation, calculated from semantic clustering of objects in MapAnything, with distance estimated from a monocular image, rather than relying on odometry. All sensors include only a low-cost monocular camera — it does not use stereo, RGB-D, or LiDAR; the accuracy mainly comes from algorithm design. The methodology is validated over sixteen independent sessions at three distances ($3\,\mathrm{m}$, $5\,\mathrm{m}$, $7\,\mathrm{m}$), showing stable and consistent results.

II. Related works

The slam systems such as ORB-SLAM [1] optimize camera pose and structure through bundle adjustment and loop closure; the semantic SLAM systems such as SemanticFusion [2], Kimera [3] extend this approach by attaching class labels to the geometric map in real time. This requires online pose optimization [4], which is not suitable for the limited computational resources of edge devices used in this research. (Section III-B).

Detecting objects on YOLO [5], [6]; other methods such as Faster R-CNN [7] or DETR [8] can achieve higher accuracy, but are heavier. With 3D reconstruction using monocular camera images, these direct inference models such as (feed-forward) MapAnything [9], Depth Anything [10], and MiDaS [11] replace the traditional pipeline SfM/MVS with a single inference pass, without multiview camera calibration.

To transform between intrinsic space of the reconstruction model and real-world space from odometry, the Umeyama algorithm [12] is classical method for registering fixed point with unknown scale, it is often combined ICP [13] for local alignment; DBSCAN [14] clusters points in the map without requiring to know the number of clusters in advance, making it suitable for separating semantic classes from label noise (Algorithm 1).

Unlike the approaches above, the method used in the paper does not optimize pose online or perform triangulation for each frame; it separates orientation signals and distance (single-image) from pose odometry for each frame — it is suitable for the post-processing constraint and low-cost hardware of the Hawkbot base.

III. Methodology

A. Hardware

The entire system is built from popular hardware, low-cost (Table I) — Not using camera stereo, RGB-D sensor,

TABLE I: System configuration

Components	Configurations
Robot	Hawkbot, ROS2 [15] Jazzy (container `hawkbot-bringup`)
Camera	ESP32-CAM, QVGA resolution (320 × 240 px)
Odometry	`nav_msgs/Odometry`, topic `/odom`
Processing	remote GPU server, connected through a local network
Detection	YOLO11n [5], [6] (nano), trained on the COCO-80 dataset
3D reconstruction	MapAnything [9] (`facebook/map-anything`)

or LiDAR. This choice makes accuracy mainly depend on algorithm design, rather than on sensor quality.

Camera intrinsic matrix K (Calibration using checkerboard pattern, 74/75 valid images, with an average reprojection error of 0.12 pixel):

$$K = \begin{bmatrix} 428.72 & 0 & 148.61 \\ 0 & 427.60 & 126.02 \\ 0 & 0 & 1 \end{bmatrix}$$

B. Current hardware limitations

Camera resolution. The camera ESP32-CAM is configured at QVGA to ensure sufficient bandwidth for image transmission and a stable frame rate. This low resolution significantly limits the angular localization accuracy of any method based on pixel positions. When evaluating the use of the localization marker ArUco [16] 15 cm(Section V-B), a one-degree error in estimating the marker angle, at a distance of more than two meters, can correspond to a real-world positional error of up to several tens of centimeters.

Computational capacity. ESP32-CAM does not perform computer vision processing locally; all Yolo and MapAnything inference runs on a remote GPU server. The current pipeline operates offline in post-processing rather than as an online SLAM system; adding an online correction mechanism, such as loop closure and pose-graph optimization, requires computational resources the current platform cannot provide.

These two constraints motivate the software-only approach of Section III-D, rather than a hardware upgrade (Section V-C).

C. Pipeline Architecture

Three main processing block of pipeline: **(i) 3D reconstruction** using MapAnything (S3), **(ii) Attaching semantic class label YOLO on 3D map** to have semantic point cloud (S4), and **(iii) Estimating single-image distance** (S7) — two signals (ii) and (iii) then is combined without triangulation through many frames.

S1. Collection: Robot moves independently; the camera continuously captures images (∼1 frame/second), recording localization/ odometry orientation for each frame into `poses.json`.

S2. Priority frame selection: Running YOLO on the overall original frame, selecting up to 130 frames containing the target object, and uniformly sampling across each visit.

S3. 3D reconstruction: Running MapAnything [9] on selected frames, reconstructing a pure geometric point cloud and the intrinsic camera trajectory.

S4. Attaching semantic class label (YOLO → 3D): Running YOLO on each frame, attaching COCO class label for each 3D point in bounding box (Algorithm 1), creates *semantic point cloud*.

S5. Global fit: Umeyama fit between MapAnything trajectory and odometry trajectory → $(s, R, \mathbf{t})$, calculated once for overall session.

S6. Object orientation: Filtering point belonging to the target class label, DBSCAN clustering [14], Selecting max cluster center; apply global fit to inference the orientation.

S7. Object distance: on the only frame, Single-image estimation from the YOLO bounding box height, real height, and camera focal length (§ III-D).

S8. Combining: localization = reference camera localization + orientation × distance.

Algorithm 1 describes the mechanism for attaching semantic class labels for 3D points at S4 — basic for determining target clusters at S6.

Algorithm 1 Attaching semantic class labels for 3D point cloud

Require: Selected frame $\{I_1, \ldots, I_n\}$; MapAnything $\mathcal{M}$; YOLO $\mathcal{Y}$
Ensure: labeled Point cloud $\mathcal{P} = \{(\mathbf{x}_j, l_j)\}$

1: $\mathcal{P} \leftarrow \emptyset$
2: **for** $i \leftarrow 1$ **to** n **do**
3: $(D_i, K_i, T_i, M_i) \leftarrow \mathcal{M}(I_i)$ ▷ depth, K, pose, mask
4: $\mathbf{P}_i \leftarrow \textsc{DepthToWorld}(D_i, K_i, T_i)$
5: $B_i \leftarrow \mathcal{Y}(I_i)$ ▷ $(l, \text{bbox}, \text{conf})$, Filtering by conf
6: **for** each pixel (u, v), $M_i(u, v) = \text{true}$ **do**
7: $l \leftarrow -1$ ▷ default: background
8: **for** each $(l', \text{bbox}, \text{conf}) \in B_i$ **do**
9: **if** $(u, v) \in$ bbox **then**
10: $l \leftarrow l'$; **break**
11: **end if**
12: **end for**
13: $\mathcal{P} \leftarrow \mathcal{P} \cup \{(\mathbf{P}_i(u, v),\ l)\}$
14: **end for**
15: **end for**
16: **return** $\mathcal{P}$

Limitations of labeling mechanism. The check at line 8 only determines whether a pixel is inside the 2D YOLO *bounding box*, not semantic segmentation at the pixel level. Background points or other objects falling inside the bounding box may also be incorrectly labeled. This is why the DBSCAN step at S6 is necessary: the densest point cluster — corresponding to the actual object observed consistently from multiple views — is separated from scattered noise points.

D. Formula

Global fit (Umeyama fit) [12]. Let $\{\mathbf{q}_k\}$ denote camera positions from MapAnything (projected onto the horizontal plane) and $\{\mathbf{p}_k\}$ denote odometry positions, respectively. The

Umeyama algorithm finds $s \in \mathbb{R}$, $R \in SO(2)$, and $\mathbf{t}$ that minimize:

$$(s^*, R^*, \mathbf{t}^*) = \arg\min_{s,R,\mathbf{t}} \sum_k \|\mathbf{p}_k - (sR\mathbf{q}_k + \mathbf{t})\|^2 \quad (1)$$

It is solved using SVD on the cross-covariance matrix between the two centered point sets. The point $\mathbf{x}_{\text{raw}}$ in the MapAnything is transformed into real-world space as: $\mathbf{x}_{\text{real}} = s^* R^* \mathbf{x}_{\text{raw}} + \mathbf{t}^*$.

Orientation. With $\mathbf{c}_{\text{raw}}$ is centroid of the largest semantic point cluster and $\mathbf{o}$ is reference camera position:

$$\hat{\mathbf{b}} = \frac{(s^* R^* \mathbf{c}_{\text{raw}} + \mathbf{t}^*) - \mathbf{o}}{\|(s^* R^* \mathbf{c}_{\text{raw}} + \mathbf{t}^*) - \mathbf{o}\|} \quad (2)$$

Single-image distance. With bbox height h_{px}, real height H, focal length f_y:

$$d = \frac{f_y \cdot H}{h_{\text{px}}}, \qquad \mathbf{x}^* = \mathbf{o} + d\,\hat{\mathbf{b}} \quad (3)$$

Distance-size ambiguity. Formula (3) assumes a known height H for the target class. Two objects of the same class with different true heights $H_1 \neq H_2$ at distances d_1, d_2 satisfying

$$\frac{H_1}{d_1} = \frac{H_2}{d_2} \left(= \frac{h_{\text{px}}}{f_y}\right), \quad (4)$$

produce the same h_{px} and cannot be distinguished by Eq. (3) alone — a limitation inherent to monocular imaging that MapAnything's depth inference does not resolve either (Section V-A). DBSCAN clustering at S6 mitigates confusion with transient false detections by favoring the densest, most persistent cluster, but not confusion between two co-located instances of the same class; the method assumes a single target instance, consistent with the current setup (Section V-C).

None of quantities above depend on the pose odometry of a single frame, except for $\mathbf{o}$ (reference results) — the alignment uses entire trajectory, while the distance estimation does not use odometry Figure 1 is a visual overview of entire localization pipeline that described above, include three blocks (i) 3D reconstruction, (ii) Attaching YOLO semantic labels, and (iii) distance estimation.

IV. Experimental Results

Each capture session follows the B1–B8 procedure of Section III-D. The method is validated over six independent sessions at $3.00\,\text{m}$ (measured with a measuring tape), the robot observing the plant pot from a different trajectory each time. To evaluate stability at greater range, the process is repeated at $5\,\text{m}$ and $7\,\text{m}$, five sessions each, combined with the $3\,\text{m}$ results in Table II and analyzed separately in Section IV-D. Figure 2 illustrates the experimental environment from a full-room scan (Section IV-C).

A. Quantitative results

Table II summarizes the results per session along with three quality indicators: YOLO detection confidence at the distance-estimation frame, the corresponding bounding-box height, and the number of points in the largest semantic cluster used for orientation.

Six sessions show errors in the range 2–9%; no session exceeds $0.26\,\text{m}$. YOLO confidence varies considerably (0.63–0.90) while semantic cluster size stays high (>29 700/30 000 points, over 99% of sampled points), showing that MapAnything reconstructs a contiguous, largely unfragmented point cluster regardless of detection confidence.

Correlation between input quality and error. The Pearson correlation is $r = 0.34$ (YOLO confidence) and $r = 0.20$ (cluster size) against absolute error — both weak and statistically insignificant at $n = 6$, consistent with Section V-A: accuracy comes from separating the two signals from per-frame pose odometry, not from high-confidence detection or large clusters.

B. Visualization on the point map

Figure 6 illustrates localized object positions provided with the robot trajectory on the point cloud map of six captured sessions. Figure 5 visualizes the distribution of estimated distances of six sessions around the ground truth measurements $3.00\,\text{m}$, showing that the estimates converge around the reference line within the range ± 1 standard deviation. The results of six sessions show average errors of about 6%, not clearly depending on trajectory shape (Figure 6) or the 2D detection confidence (Table II) — consistent with the argument presented in Section V-A. Figure 4 Comparison of the real-world experimental environment with a YOLO11n frame detecting the plant pot.

C. Extension: Full-room semantic map scanning

To evaluate scalability of the pipeline beyond the single target localization scenario, we move the robot around the room (239 frames, instead of the 130 frames focused on a single object as in Section IV) and run the entire pipeline again for S1–S8 unlimited target semantic classes. The result is a 3D map with 38.8 million points, while YOLO11n attaches 12 different COCO labels. Figure 2 compares a real-world photograph of the room with the top-down view of the point map.

The reconstructed map accurately preserves the overall shape and layout of the furniture (tables, chairs, cabinets, and the plant pot at their corresponding relative positions). However, due to the QVGA camera combined with the YOLO11n nano model, some detected labels are false positives—for example, "car", "airplane", and "traffic light", which are largely absent from the actual office environment—consistent with the hardware limitations discussed in Section III-B. These results demonstrate that the pipeline generalizes well to full-room mapping, although the reliability of semantic labeling remains constrained by the current sensing hardware.

D. Distance-dependent accuracy evaluation

The six capture sessions in Section IV were all conducted at $3.00\,\text{m}$. To evaluate whether the method remains accurate at greater target distances, we repeated the B1–B8 procedure, keeping $H = 1.25\,\text{m}$ and using the same target plant pot, at

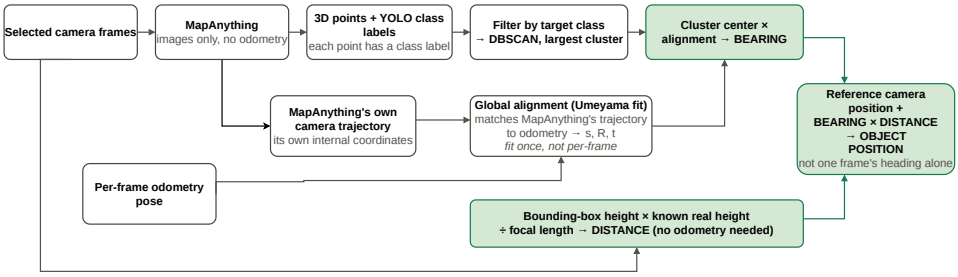
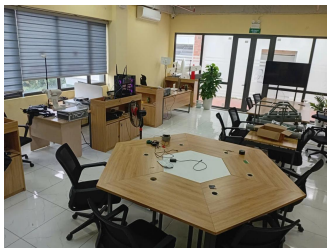


Fig. 1: Object localization pipeline combining bearing estimated from MapAnything semantic point clusters with monocular distance estimation.



(a) Real-world captured image

(b) Point map (top view, aligned with the direction of the real-world image))

Fig. 2: Full-room scan: real-world image (left) and 3D point map reconstructed from 239 frames (right). Yellow marker: centroid of the largest point cluster; red line: robot trajectory.

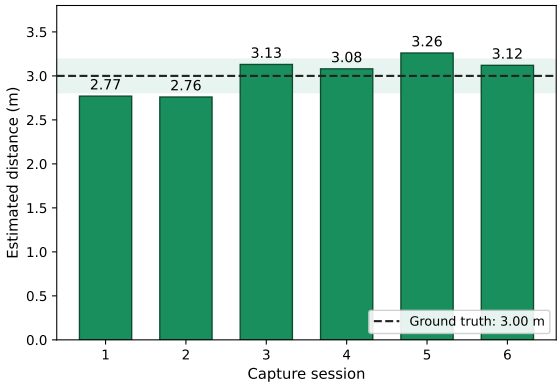


Fig. 3: Estimated distances across six capture sessions, compared with the ground-truth distance of $3.00\,\mathrm{m}$ (dashed line). Shaded region: ± 1 standard deviation around the mean.

(a) Experimental environment

(b) YOLO11n plant pot detection

Fig. 4: Real-world experimental environment (left): the Hawkbot robot positioned in the middle of the floor in an office space, with the plant pot as the target object. YOLO11n detection on an ESP32-CAM frame (right), with a confidence score of 0.75.

two additional distances: five sessions at $5\,\mathrm{m}$ and five sessions at $7\,\mathrm{m}$, with the results summarized in Table II.

The three result groups reveal two distinct error patterns. At $3\,\mathrm{m}$ and $5\,\mathrm{m}$, the mean error nearly cancels ($+0.6\%$ and -1.5%), but the standard deviations are relatively large ($0.19\,\mathrm{m}$, $0.22\,\mathrm{m}$), with individual sessions deviating up to $\pm 10\%$ — characteristic of random noise. At $7\,\mathrm{m}$, in contrast, the standard deviation is very small ($0.07\,\mathrm{m}$, less than one-third of the two shorter ranges), yet all five sessions consistently show a -11% bias, characteristic of a systematic error. Because $d = f_y H / h_{\mathrm{px}}$ gives a proportional relationship between d and $1/h_{\mathrm{px}}$, a constant relative error in f_y (camera calibration) or H (assumed plant-pot height) produces exactly the observed nearly-constant bias. Recalibrating f_y or measuring H more accurately is therefore a direct mitigation (Section V-C).

V. Discussion

A. Why Global Fitting Is More Stable Than Per-Frame Triangulation

Both signals used by the method avoid dependence on the odometry pose of any single frame. The Umeyama alignment uses the entire camera trajectory, reducing the effect of local pose noise, while the monocular distance depends only on the object's image size and the calibrated camera intrinsics.

TABLE II: Plant localization results at three real-world distances (3 m: six capture sessions; 5 m and 7 m: five sessions each), together with input quality indicators

Real-world distance	Capture session	Estimated distance	Error w.r.t. ground truth	YOLO confidence	BBox height (px)	Semantic cluster points
3 m	1	2.77 m	−7.6%	0.63	192.9	29 841
	2	2.76 m	−8.1%	0.90	193.9	29 992
	3	3.13 m	+4.2%	0.78	170.9	29 974
	4	3.08 m	+2.6%	0.65	173.6	29 773
	5	3.26 m	+8.5%	0.76	164.2	29 913
	6	3.12 m	+3.9%	0.76	171.4	30 000
	Mean	3.02 m	Std. dev. 0.19 m, mean error +0.6% (absolute mean 6.2%)			
5 m	1	4.49 m	−10.2%	0.81	119.1	29 606
	2	5.07 m	+1.4%	0.74	105.4	29 984
	3	5.00 m	+0.0%	0.85	106.9	25 826
	4	5.06 m	+1.2%	0.82	105.6	25 932
	5	5.01 m	+0.3%	0.75	106.6	22 543
	Mean	4.93 m	Std. dev. 0.22 m, mean error −1.5% (absolute mean 2.6%)			
7 m	1	6.21 m	−11.3%	0.85	86.1	29 742
	2	6.22 m	−11.1%	0.51	85.9	29 588
	3	6.17 m	−11.9%	0.73	86.6	30 000
	4	6.34 m	−9.4%	0.71	84.3	29 662
	5	6.13 m	−12.4%	0.82	87.2	29 933
	Mean	6.21 m	Std. dev. 0.07 m, mean error −11.2%			

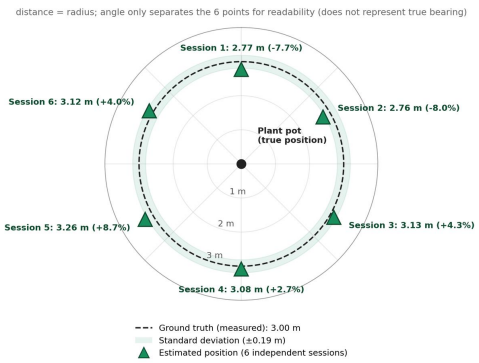


Fig. 5: Distribution of estimated distances across six independent capture sessions around the ground-truth distance of 3.00 m (dashed circle). Green triangles denote the estimated distance for each session, the black dot at the center indicates the true plant-pot position, and the shaded region represents the ±1 standard deviation range (0.19 m). The angular positions are used only to separate the points for readability and do not represent actual bearing directions.

Empirical validation. To evaluate whether the direct 3D position from MapAnything could replace the monocular distance, we compute $\|(s^* R^* \mathbf{c}_{\text{raw}} + \mathbf{t}^*) - \mathbf{o}\|$ for all sixteen sessions without using Eq. (3). The results are compared with the proposed method in Table III.

Directly using the 3D position from MapAnything produces substantially larger errors than the monocular formulation, with one case reaching nearly −80%. The Umeyama alignment fits a single (s, R, t) over the whole trajectory and cannot guarantee point-level accuracy, especially when rotational segments distort the geometry. Moreover, MapAnything's depth estimation does not resolve the distance–size ambiguity — it merely shifts the metric reference from the known H to odometry. We therefore use the 3D position only for bearing, which is less sensitive to global scale errors, while monocular geometry provides the absolute distance.

TABLE III: Mean error: monocular distance vs. direct MapAnything 3D position

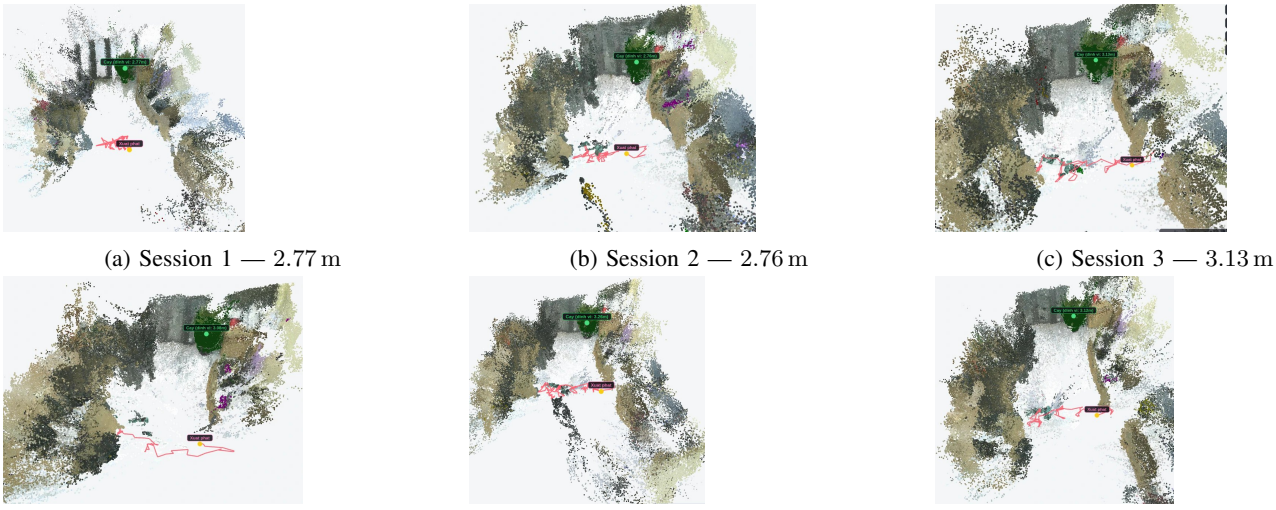
Distance Ground truth	Single-image (in paper)	3D position (direct)
3 m	+0.6%	−21.9%
5 m	−1.5%	−22.2%
7 m	−11.2%	−38.4%

B. Hardware Alternatives Tested

- **ArUco:** A 15 cm marker gave `solvePnP` errors of 0.3 m to 0.7 m beyond 2 m or under oblique views — as large as the drift it was meant to correct.
- **EKF (`/odometry/filtered`):** covariance diverged to $\sim 10^{22}$, unusable without further tuning.

C. Limitations

The study is based on sixteen capture sessions (six at 3 m, five at 5 m, and five at 7 m) in a single room with one target type (plant pot), which limits the assessment of generalization. The monocular distance estimate requires the



(a) Session 1 — 2.77 m (b) Session 2 — 2.76 m (c) Session 3 — 3.13 m

(d) Session 4 — 3.08 m (e) Session 5 — 3.26 m (f) Session 6 — 3.12 m

Fig. 6: Localized plant-pot position (green marker) and robot trajectory (red line, starting point marked) on the MapAnything point-cloud map for six independent capture sessions.

true object height to be known, making it less suitable for objects with varying dimensions. The systematic -11% bias at $7\,\mathrm{m}$ suggests that H or f_y should be calibrated more accurately for longer ranges. Global alignment also depends on sufficient trajectory diversity; short or nearly straight trajectories may reduce its reliability. Future work should evaluate more object types, capture sessions, and environments, while higher-resolution cameras and greater edge-computing capacity could enable more robust solutions.

VI. Conclusion

We presented an absolute object-localization method in a semantic 3D map for an autonomous delivery robot, combining bearing estimated from semantic point clusters through global trajectory alignment with monocular distance from a single frame, without relying on odometry. No individual step requires an accurate pose from a single frame. Across sixteen capture sessions at $3\,\mathrm{m}$, $5\,\mathrm{m}$, and $7\,\mathrm{m}$, the method achieved mean errors below 2% at the two shorter ranges, while showing a systematic bias of approximately -11% at $7\,\mathrm{m}$. This indicates the need for more accurate calibration of the assumed object height or camera focal length when extending the operating range. Separating bearing estimation from distance estimation provides an effective localization design for low-cost robotic platforms.

References

[1] R. Mur-Artal, J. M. M. Montiel, and J. D. Tardós, "Orb-slam: A versatile and accurate monocular slam system," *IEEE Transactions on Robotics*, vol. 31, no. 5, pp. 1147–1163, Oct 2015.

[2] J. McCormac, A. Handa, A. Davison, and S. Leutenegger, "Semanticfusion: Dense 3d semantic mapping with convolutional neural networks," in *2017 IEEE International Conference on Robotics and Automation (ICRA)*, May 2017, pp. 4628–4635.

[3] A. Rosinol, M. Abate, Y. Chang, and L. Carlone, "Kimera: an open-source library for real-time metric-semantic localization and mapping," in *2020 IEEE International Conference on Robotics and Automation (ICRA)*, May 2020, pp. 1689–1696.

[4] C. Cadena, L. Carlone, H. Carrillo, Y. Latif, D. Scaramuzza, J. Neira, I. Reid, and J. J. Leonard, "Past, present, and future of simultaneous localization and mapping: Toward the robust-perception age," *IEEE Transactions on Robotics*, vol. 32, no. 6, pp. 1309–1332, Dec 2016.

[5] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2016, pp. 779–788.

[6] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics yolo11," 2024. [Online]. Available: https://github.com/ultralytics/ultralytics

[7] S. Ren, K. He, R. Girshick, and J. Sun, "Faster r-cnn: Towards real-time object detection with region proposal networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 39, no. 6, pp. 1137–1149, June 2017.

[8] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. ECCV*, 2020.

[9] Meta AI (FAIR), "MapAnything: Universal feed-forward 3d reconstruction," 2025. [Online]. Available: https://huggingface.co/facebook/map-anything

[10] L. Yang, B. Kang, Z. Huang, X. Xu, J. Feng, and H. Zhao, "Depth anything: Unleashing the power of large-scale unlabeled data," in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2024, pp. 10 371–10 381.

[11] R. Ranftl, K. Lasinger, D. Hafner, K. Schindler, and V. Koltun, "Towards robust monocular depth estimation: Mixing datasets for zero-shot cross-dataset transfer," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 3, pp. 1623–1637, March 2022.

[12] S. Umeyama, "Least-squares estimation of transformation parameters between two point patterns," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 13, no. 4, pp. 376–380, April 1991.

[13] P. Besl and N. D. McKay, "A method for registration of 3-d shapes," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 14, no. 2, pp. 239–256, Feb 1992.

[14] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proc. KDD*, 1996.

[15] M. Quigley, K. Conley, B. Gerkey, J. Faust, T. Foote, J. Leibs, E. Wheeler, and A. Y. Ng, "ROS: an open-source robot operating system," in *ICRA Workshop on Open Source Software*, 2009.

[16] S. Garrido-Jurado, R. Muñoz-Salinas, F. Madrid-Cuevas, and M. Marín-Jiménez, "Automatic generation and detection of highly reliable fiducial markers under occlusion," *Pattern Recognition*, vol. 47, no. 6, pp. 2280–2292, 2014. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0031320314000235